Statistics Formula Sheet

Every formula from chapters 1–13 of these notes, each with a sketch of what it means. Grey numbers (e.g. **3.11**) point to the lesson note.

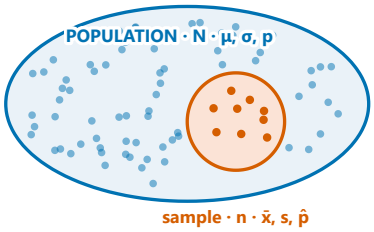
In the sketches:
population · H_0 · main quantity
sample · H_a · what is asked
tail · rejection region

1 Introduction to statistics

Population vs sample

1.2 · 1.4

	size	mean	SD	var.	prop.
population → **parameter**	N	μ	σ	σ^2	p
sample → **statistic**	n	$\bar{x}$	s	s^2	$\hat{p}$



Descriptive
summarize the data you have: tables, graphs, x̄, median, s

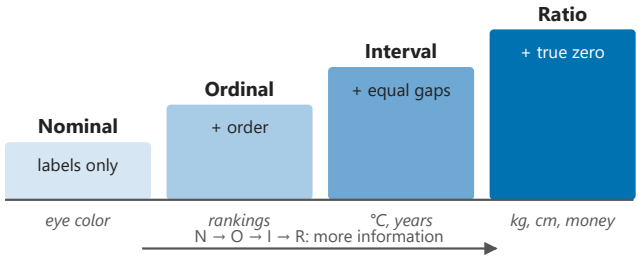
Inferential
generalize sample → population with a stated confidence

We compute statistics ($\bar{x}, s, \hat{p}$) to **estimate** unknown parameters (μ, σ, p). Bigger samples → more precise inferences.

Types of data & scales of measurement

1.3 · 1.5 · 6.5

- **Quantitative** (numbers): **discrete** = counted, whole values · **continuous** = measured, any value in a range.
- **Qualitative** (categories, words): colors, tastes, opinions.



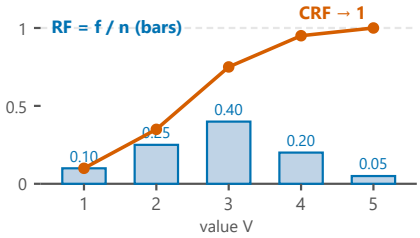
Ratios like "twice as much" only make sense on a **ratio** scale (true zero): 20 kg is twice 10 kg, but 20 °C is not twice as hot as 10 °C.

2 Graphs and tables

Frequency tables

2.1 · 2.2 · 2.3

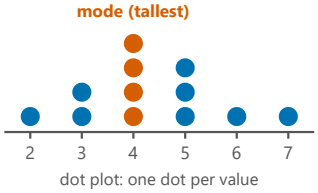
total	$n = \sum f$	relative freq.	$\mathrm{RF} = \dfrac{f}{n}$ (sums to 1)
cumulative RF	CRF = running sum of RF; never decreases, ends at **1.00** → "share at or below"		
mean from table	$\bar{x} = \dfrac{\sum V f}{n}$	median	position $\frac{n+1}{2}$ via running f



V	f	RF	CRF
1	2	.10	.10
2	5	.25	.35
3	8	.40	.75
4	4	.20	.95
5	1	.05	1.00
Σ	**20**	**1.00**	

Dot plots, line plots, stem-and-leaf

2.4 – 2.7



stem | leaf

4 | 2 7
5 | 1 3 8
6 | 0 4 4 9
7 | 5

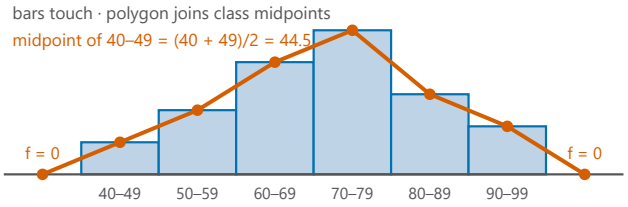
key: 5 | 3 = 53

- Height of a column = **frequency**; mode = tallest column; total amount = $\sum(\text{value} \times \text{frequency})$.
- "**at least** k" = $x \geq k$, "**at most** k" = $x \leq k$ — both include k.
- Stem = leading digit(s), leaf = last digit, sorted, no empty stems skipped, **always a key**. Side-by-side: left leaves run in reverse.

Histogram & frequency polygon

2.8 · 2.9

$$\text{class midpoint} = \frac{\text{lower bound} + \text{upper bound}}{2}$$



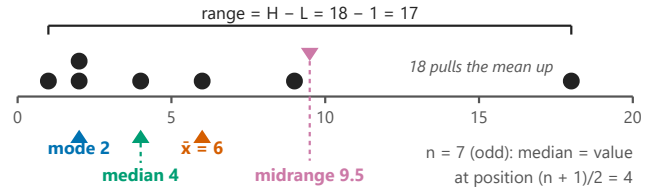
Equal-width classes. Histogram bars **touch** (bar-graph bars don't). For the polygon, add a point with $f = 0$ one class width beyond each end. Long right tail → **skewed right**.

3 Central tendency and variation

Mean, median, mode, range

3.2 – 3.5 · 3.9

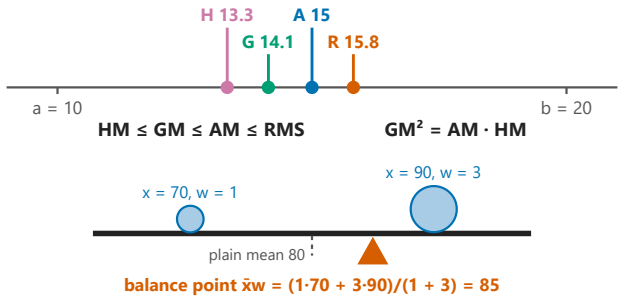
mean	$\bar{x} = \frac{\sum x}{n}$ (sample) $\mu = \frac{\sum x}{N}$ (population)
median (sorted!)	n odd: $a_{\frac{n+1}{2}}$ n even: $\frac{a_{n/2} + a_{n/2+1}}{2}$
mode	most frequent value (can be none, or several)
range · midrange	$H - L$ · $\frac{H+L}{2}$ · coeff. of range $\frac{H-L}{H+L}$
missing value	$n \times \text{target mean} - \sum(\text{known values})$



Four means & the weighted mean

3.1 · 3.6 · 3.7

mean	n values	two values a, b
arithmetic A	$\frac{x_1 + \cdots + x_n}{n}$	$\frac{a+b}{2}$
geometric G	$(x_1 x_2 \cdots x_n)^{1/n}$	$\sqrt{ab}$
harmonic H	$\frac{n}{\sum 1/x}$	$\frac{2ab}{a+b}$
root mean square R	$\sqrt{\frac{\sum x^2}{n}}$	$\sqrt{\frac{a^2+b^2}{2}}$
weighted	$\bar{x}_w = \frac{\sum wx}{\sum w}$ (x = what you average, w = how much each counts)	

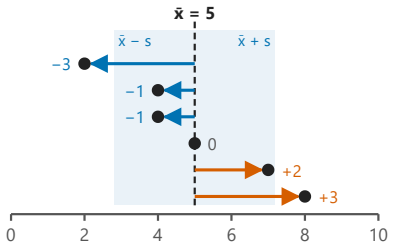


Harmonic = average speed over equal distances. **Weighted** = grades, GPA, mixtures; pulled toward the heaviest weights.

Variance, standard deviation, MAD

3.11 · 3.13 · 3.18 · 3.19

	sample (÷ $n-1$)	population (÷ N)
variance	$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$	$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$
standard deviation	$s = \sqrt{s^2}$	$\sigma = \sqrt{\sigma^2}$
mean abs. deviation	$\text{MAD} = \frac{\sum\lvert x-\bar{x}\rvert}{n}$ always	signed: $\frac{\sum(x-\bar{x})}{n} = 0$



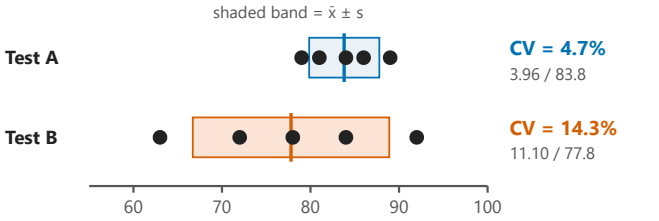
data 2, 4, 4, 5, 7, 8
Σ(x − x̄) = 0 (always)
squares 9+1+1+0+4+9
Σ(x − x̄)² = 24
s² = 24/(6 − 1) = 4.8
s = √4.8 ≈ 2.19
MAD = 10/6 ≈ 1.67

SD = typical distance from the mean; variance is in **squared** units. Excel: `STDEV.S` / `VAR.S` (sample), `STDEV.P` / `VAR.P` (population).

Coefficient of variation & quick uncertainty

3.10 · 3.12

coefficient of variation	$\text{CV} = \frac{s}{\bar{x}} \times 100\%$ (population: $\frac{\sigma}{\mu} \times 100\%$)
quick uncertainty	report $\bar{x} \pm u$, $u = \frac{\text{range}}{2}$ rounded to 1 significant figure

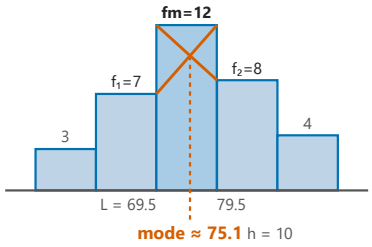


CV has **no units** → use it to compare spread between data sets with different means or units. Test B varies 3× more relative to its mean.

Grouped data

3.14 – 3.17

mean	$\bar{x} \approx \frac{\sum fm}{\sum f}$, m = class midpoint
SD	$s \approx \sqrt{\frac{\sum f(m-\bar{x})^2}{n-1}}$, $n = \sum f$
median	$L + \left(\frac{\frac{n}{2} - C}{f}\right) h$ (C = cum. freq. **before** median class)
mode	$L + h\left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right)$ (f_1 before, f_2 after)



classes 50–59 … 90–99
n = Σf = 34
cum. f: 3, 10, 22, …
n/2 = 17 → median class
70–79, C = 10, f = 12
median = 69.5 + (17−10)/12·10
≈ 75.3
x̄ ≈ Σfm / Σf ≈ 75.4

L = lower **boundary** (69.5, not 70) and h = boundary width (10, not 9). Results are estimates.

Statistics in Excel

3.8

basics	`SUM` `COUNT` `AVERAGE` `MEDIAN` `MODE` `MIN` `MAX`
spread	`STDEV.S` `VAR.S` (sample) · `STDEV.P` `VAR.P` (population)
position	`QUARTILE(range, 0…4)` · `PERCENTILE(range, k/100)` · decile D_k = `PERCENTILE(range, k/10)`
regression	`SLOPE` `INTERCEPT` `CORREL` · tables: `NORM.S.DIST`, `T.INV`, `CHISQ.INV.RT`

Excel's quartile rule can differ slightly from the hand method — several conventions exist.

4 Measures of position

Quartiles, deciles, percentiles

4.1 · 4.3

position of P_k	$L_k = \frac{k}{100}(n+1)$ quartile: $\frac{k(n+1)}{4}$
percentile of value v	$\frac{x + 0.5\,y}{n} \times 100$ (x below v, y equal to v)
same thing	$Q_1 = P_{25},\ Q_2 = D_5 = P_{50},\ Q_3 = P_{75}$

sorted data, n = 11

1	2	3	4	5	6	7	8	9	10	11
2	4	**5**	7	8	**10**	12	13	**15**	18	20

$Q_1 = 5$ $Q_2 = 10$ (median) $Q_3 = 15$

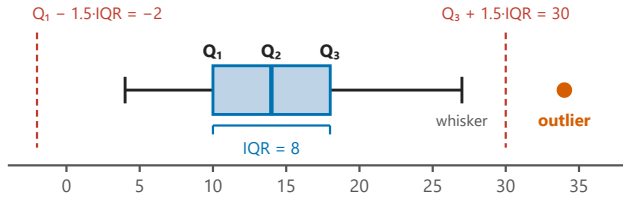
position of Qk = k(n + 1)/4 → 3, 6, 9 each quarter holds 25%

> Position not a whole number → average the two neighbours (or interpolate: 10.2 → $a_{10} + 0.2(a_{11} - a_{10})$).

IQR, box plots & outliers

4.2 · 4.3 · 4.4

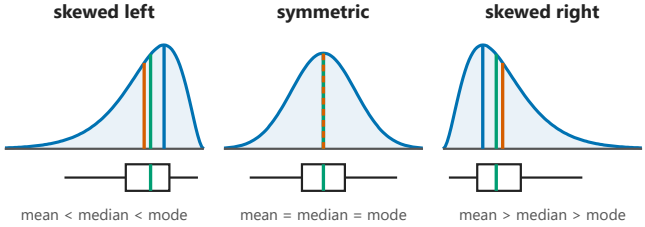
IQR · quartile deviation	$\mathrm{IQR} = Q_3 - Q_1$, $\mathrm{QD} = \frac{Q_3 - Q_1}{2}$
coeff. of QD (no units)	$\mathrm{CQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$
outlier fences	$Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$



> Box = Q_1 to Q_3, line = median; whiskers reach the most extreme **non-outliers**; values beyond the fences are dots.

Shape & skewness

4.5 · 1.1



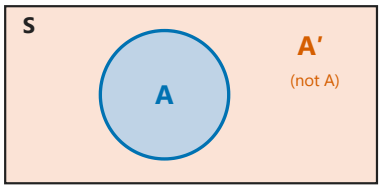
> Lines: **mode**, **median**, **mean**. The mean is pulled toward the tail. On a box plot the **longer side** points the way of the skew. One / two / many peaks = unimodal / bimodal / multimodal.

5 Probability

Basic probability & complements

5.1 · 5.3 · 5.8

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{favorable}}{\text{total}}, \qquad 0 \le P(A) \le 1$$
$$P(A') = 1 - P(A)$$



P(A) + P(A') = 1
P(A') = 1 − P(A)

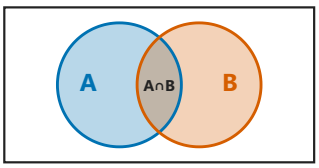
"at least one"
= 1 − P(none)

> n fair coin flips in one given order: $(1/2)^n$. "In any order" → count every arrangement that works.

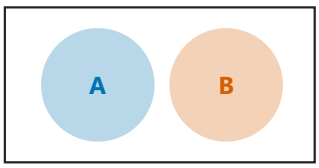
Addition rule — "or"

5.2 · 5.6 · 5.7

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
mutually exclusive: $P(A \cup B) = P(A) + P(B)$



overlap counted twice → subtract once



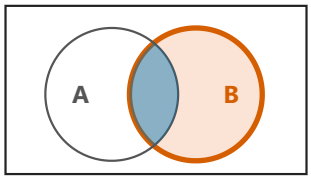
mutually exclusive: P(A∩B) = 0

> **or** → ∪ → add (minus the overlap). A probability above 1 means an overlap was counted twice.

Conditional probability & multiplication — "and"

5.4 · 5.5 · 5.7 · 5.8

conditional	$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$
multiplication	$P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$
independent	$P(A \cap B) = P(A)\,P(B)$ (with replacement)
chain (dependent)	$P(A_1 \cap B_2 \cap C_3) = P(A_1)\,P(B_2 \mid A_1)\,P(C_3 \mid A_1 \cap B_2)$



given B: B becomes the whole space

	A	A′	total
B	15	15	30
B′	25	45	70
total	40	60	100

P(A | B) = 15/30 = 0.50
but P(A) = 40/100 = 0.40

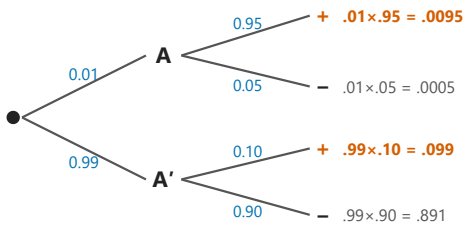
> **Independent ≠ mutually exclusive.** Independent: one doesn't affect the other. Mutually exclusive: can't both happen, $P(A \cap B) = 0$. Without replacement → update counts **and** total.

Tree diagrams & Bayes' theorem

5.10 · 5.11

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

$$P(B) = P(B \mid A)\,P(A) + P(B \mid A')\,P(A') \quad \text{(add the branches that end in } B\text{)}$$



Bayes

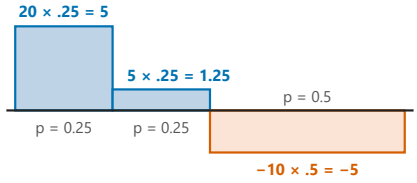
P(A | +) =
P(+|A)P(A) / P(+)
= .0095 / (.0095 + .099)
≈ 0.088
× along a branch,
+ across branches

Rare condition (1%) + 10% false positives → only ~9% of positive tests are real. Check with counts (e.g. 10,000 people).

Expected value

5.9

$$E(X) = \sum x\,P(x) = x_1p_1 + x_2p_2 + \cdots + x_np_n$$



E(X) = Σ x · P(x)
= 5 + 1.25 − 5
= 1.25 per game
(long-run average)
n games → n · E(X)

width = probability, height = value

Total expected over n trials $= n\,E(X)$. Losses go in as **negative** values; the p's add to 1. $E(X)$ is a long-run average — no single game pays exactly 1.25.

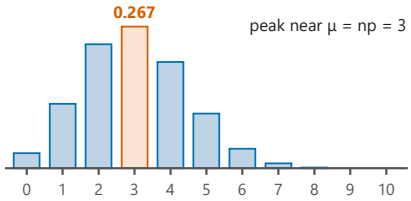
6 Discrete probability distributions

Binomial — x successes in n trials

6.1 · 6.2

$$P(X = x) = {}_nC_x\,p^x q^{n-x}, \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}, \qquad q = 1 - p$$

$$\mu = np, \qquad \sigma = \sqrt{npq}$$



n = 10, p = 0.3
P(X = 3)
= ${}_{10}C_3\,(0.3)^3\,(0.7)^7$
≈ 0.267
σ = √(npq) = √2.1
≈ 1.45

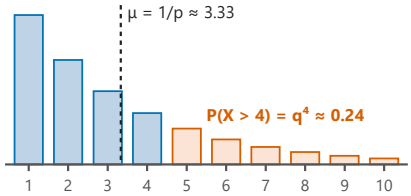
Fixed n, success/failure, independent trials, same p. "Exactly" → one term; "at most / at least / fewer than" → add terms (remember $x = 0$).

Geometric — waiting for the first success

6.1 · 6.3

on trial x	by trial x	before x	after x	x or later
$q^{x-1}p$	$1 - q^x$	$1 - q^{x-1}$	q^x	q^{x-1}

$$\mu = \frac{1}{p}, \qquad \sigma^2 = \frac{1-p}{p^2}, \qquad \sigma = \frac{\sqrt{1-p}}{p}$$

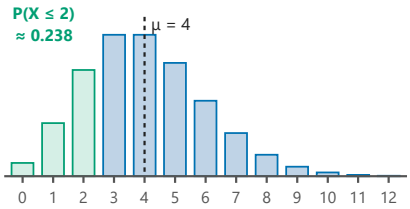


p = 0.3, q = 0.7
1st success on trial x:
P(X = x) = q^{x-1} p
P(X ≤ x) = 1 − q^x
P(X > x) = q^x
σ^2 = q / p^2

Poisson — events in an interval

6.1 · 6.4

$$P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \qquad \frac{\mu_2}{\mu_1} = \frac{t_2}{t_1}, \qquad \text{mean} = \sigma^2 = \mu, \quad \sigma = \sqrt{\mu}$$

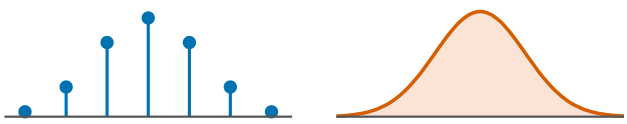


μ = 4 events / interval
P(X = x) = μ^x e^(−μ) / x!
P(X ≤ 2) = $e^{-4}(1 + 4 + 8)$
mean = variance = μ
new interval: rescale μ
$\mu_2 / \mu_1 = t_2 / t_1$

Adjust μ first whenever the question's interval differs from the given rate. "More than k" → $1 - P(X \le k)$.

Discrete vs continuous

6.5



counted → discrete
separate values, gaps between

measured → continuous
any value in a range

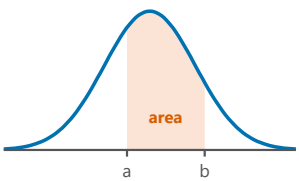
Discrete (ch. 6): binomial, geometric, Poisson — probabilities sit on single values. **Continuous** (ch. 7–8): uniform, exponential, normal — probability is **area**. Rounding turns continuous data into discrete data.

7 Continuous distributions

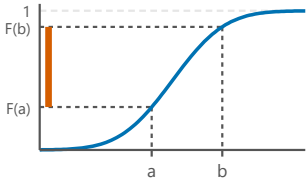
PDF & CDF rules

7.1 · 7.2

PDF $f(x)$	$f(x) \ge 0$, total area $= 1$, probability $=$ area
one value	$P(X = a) = 0 \;\Rightarrow\; P(X < a) = P(X \le a)$
CDF $F(x)$	$F(x) = P(X \le x)$ = area to the left
right · between	$P(X > x) = 1 - F(x)$, $P(a < X < b) = F(b) - F(a)$



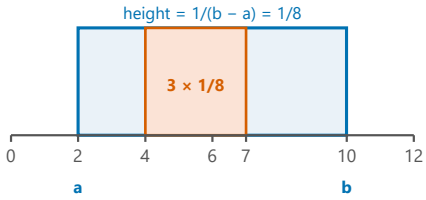
PDF: P(a < X < b) = area



CDF: F(b) − F(a), same area

Uniform $U(a, b)$ 7.3

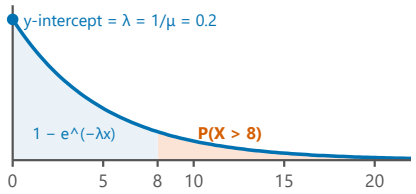
PDF · CDF	$f(x) = \frac{1}{b-a}, \qquad F(x) = \frac{x-a}{b-a}$
probability	$P = \text{base} \times \text{height} = \frac{\text{interval width}}{b-a}$
mean = median	$\mu = \frac{a+b}{2}$
spread	$\sigma^2 = \frac{(b-a)^2}{12}, \qquad \sigma = \frac{b-a}{\sqrt{12}}$
k-th percentile	$K = a + \frac{k}{100}(b-a)$ (outside $[a, b]$ the probability is 0)



U(2, 10)
P(4 < X < 7)
= width / (b − a)
= 3/8 = 0.375
μ = (a + b)/2 = 6
σ = (b − a)/√12 ≈ 2.31

Exponential — waiting times & lifetimes 7.4

rate · PDF	$\lambda = \frac{1}{\mu}, \qquad f(x) = \lambda e^{-\lambda x} \ (x \geq 0), \qquad \sigma = \mu$
left · right	$P(X < x) = 1 - e^{-\lambda x}, \qquad P(X > x) = e^{-\lambda x}$
between	$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$

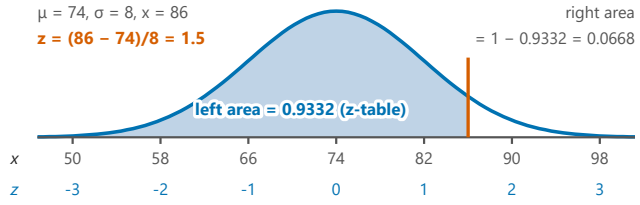


μ = 5 (e.g. years)
P(X > 8) = e^(−λx)
= e^(−1.6) ≈ 0.202
P(X < 8) ≈ 0.798
σ = μ = 1/λ

8 Normal distribution, CLT and standard error

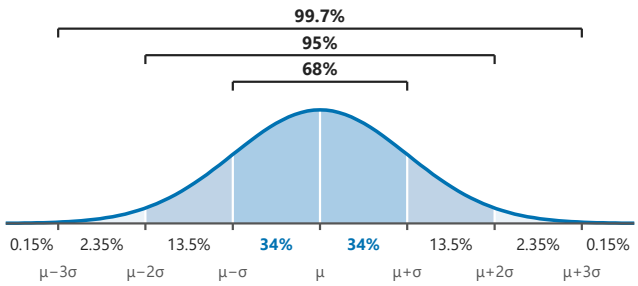
z-scores & normal probabilities 8.1 · 8.2

$$z = \frac{x-\mu}{\sigma}, \qquad x = \mu + z\sigma, \qquad f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



> z-table = **left** area. Right = 1 − left; between = difference of two lefts. Percentile: area → z → x. Count = probability × group size.

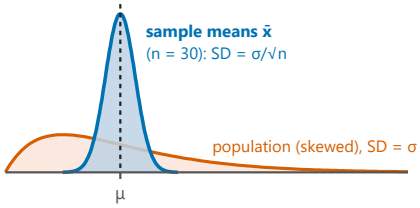
Empirical rule (68 – 95 – 99.7) 8.1 · 8.2



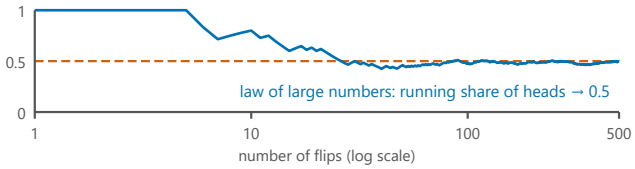
> Normal data only. Precise values: 68.27 / 95.45 / 99.73 %, per region 34.13 / 13.59 / 2.14 / 0.135 %. For any shape use Chebyshev (9.6).

Sampling distributions: LLN, CLT, standard error 8.3 · 8.4 · 8.5

question about…	distribution	z
one value x	$X \sim N(\mu, \sigma)$	$\frac{x-\mu}{\sigma}$
mean of n values	$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
sum of n values	$\sum X \sim N(n\mu, \sqrt{n}\,\sigma)$	$\frac{\sum x - n\mu}{\sqrt{n}\,\sigma}$
standard error	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ (population σ)	$\mathrm{SE} = \frac{s}{\sqrt{n}}$ (s)



μx̄ = μ
σx̄ = σ / √n
x̄ ≈ normal for n ≥ 30, whatever the shape
sum ~ N(nμ, √n σ)



> SD = spread of **individual values**; SE = spread of **sample means**. SE ∝ $1/\sqrt{n}$. The LLN says nothing about a single trial — the next flip is still 50/50.

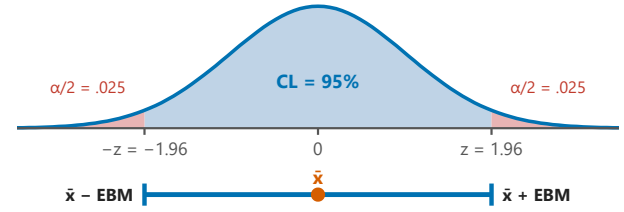
9 Confidence intervals

CI for a mean, σ known (z)

9.1 · 9.2

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}, \qquad \text{EBM} = z\frac{\sigma}{\sqrt{n}}$$

look up z from the left area $\dfrac{1+\text{CL}}{2}$



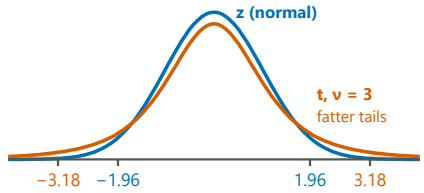
CL	90%	95%	98%	99%
left area	0.95	0.975	0.99	0.995
z	**1.645**	**1.96**	**2.33**	**2.575**

Higher confidence → bigger z → **wider**. Bigger n → **narrower**. $\alpha = 1 - \text{CL}$, split $\alpha/2$ per tail.

CI for a mean, σ unknown (t)

9.3

$$\bar{x} \pm t_{\nu,\alpha/2}\frac{s}{\sqrt{n}}, \qquad \nu = n - 1, \qquad \frac{\alpha}{2} = \frac{1-\text{CL}}{2}$$



ν = n − 1
one-tail area
α/2 = (1 − CL)/2
95%, n = 4 (ν = 3):
t = 3.182 > z = 1.96
→ wider interval

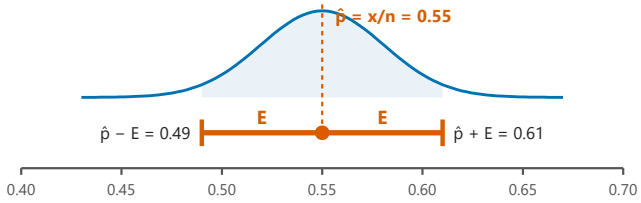
Use t when only s is known, especially $n < 30$. t-table: row ν, column = one-tail area $\alpha/2$. As ν grows, $t \to z$.

CI for a proportion

9.4

estimate $\hat{p} = \dfrac{x}{n}$, $\hat{q} = 1 - \hat{p}$ check $n\hat{p}\hat{q} \geq 10$

margin of error $E = z\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$ interval $\hat{p} - E < p < \hat{p} + E$

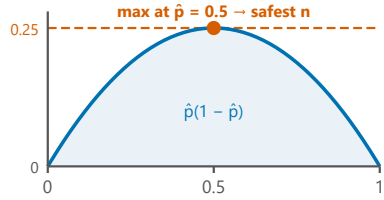


Given an interval: $\hat{p}$ = its **midpoint**, E = **half its width**. The square-root part is the standard error of $\hat{p}$.

Sample size

9.4 · 9.5

mean: $n = \dfrac{z^2\sigma^2}{E^2}$ proportion: $n = \dfrac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}$



95%, E = 0.05, p̂ = 0.5:
n = 1.96² (0.25) / 0.05²
= 384.2 → 385
(always round UP)
halve E → 4 × n

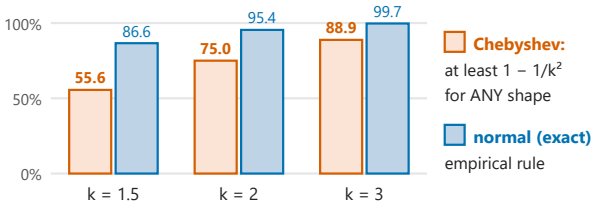
Both come from solving $E = z \times \text{SE}$ for n. No estimate of $\hat{p}$? Use **0.5** (largest n). Always round **up**. Higher confidence or smaller E → larger n; E is squared, so it matters most.

Chebyshev's theorem

9.6

at least $1 - \dfrac{1}{k^2}$ of **any** data set lies within k SDs of the mean

$$k = \frac{x - \mu}{\sigma}, \qquad k > 1$$



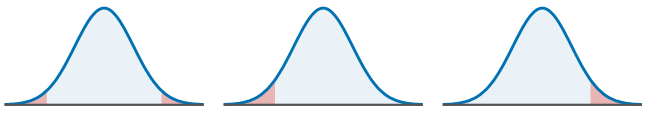
Answers are **lower bounds** — say "at least". $k = 2$ → 75%, $k = 3$ → 88.9%. Use the empirical rule only when the data are normal.

10 Hypothesis testing — one sample

Hypotheses & tails

10.1 · 10.3

wording	H_0 (equality)	H_a (strict)	test
"is different from", "is not"	$=$	$\neq$	two-tailed
"at least", "no less than"	$\geq$	$<$	left-tailed
"at most", "no more than"	$\leq$	$>$	right-tailed



H_a: ≠ two-tailed
α/2 in each tail
z = ±1.96 (α = .05)

H_a: < left-tailed
all of α on the left
z = −1.645 (α = .05)

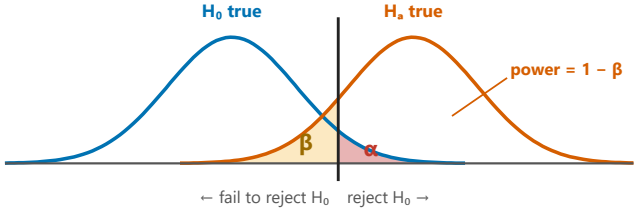
H_a: > right-tailed
all of α on the right
z = +1.645 (α = .05)

Hypotheses are about **parameters** (μ, p), never $\bar{x}$ or $\hat{p}$. Only two decisions: reject H_0, or fail to reject H_0 (never "accept"). $\alpha = 1 - c$.

Type I & Type II errors, power

10.2

decision ↓ truth →	H_0 true	H_0 false
reject H_0	**Type I** (false alarm) · α	correct · power $= 1 - \beta$
fail to reject H_0	correct · $1 - \alpha$	**Type II** (missed) · β



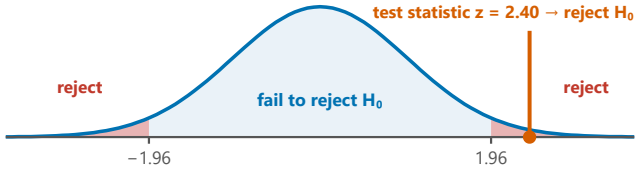
Type I = false positive, Type II = false negative. Which one is worse depends on the real-world consequences.

One-sample test statistics

10.4 · 10.5 · 10.7

$$\text{test statistic} = \frac{\text{sample value} - \text{value claimed by } H_0}{\text{standard error}}$$

mean, σ known or $n \geq 30$	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	(use s if σ unknown)
mean, σ unknown, $n < 30$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, $\text{df} = n - 1$	
proportion	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$	(p_0 from H_0 in the SE)

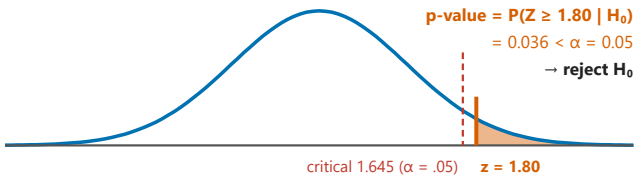


two-tailed, α = 0.05: compare the test statistic with ±1.96

α	0.10	0.05	0.01
two-tailed z	±1.645	±1.96	±2.576
one-tailed z	1.28	1.645	2.33

p-value method

10.6



two-tailed test: p-value = 2 × one-tail area

p-value = probability, **if H_0 is true**, of a result at least as extreme as the one observed. **p < α → reject H_0**; p ≥ α → fail to reject. Always gives the same decision as the critical-value method.

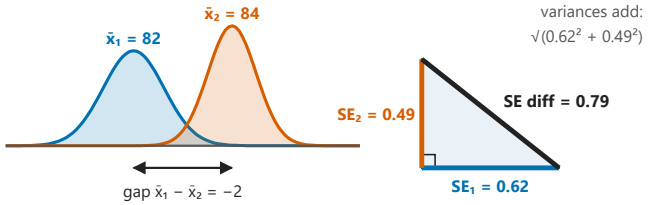
Every test, 5 steps: ① write H_0/H_a → ② pick α, tails, critical value → ③ compute the statistic → ④ compare (or p-value vs α) → ⑤ conclude in words.

11 Hypothesis testing — two samples

Difference of two means (independent)

11.1

$$t \text{ (or } z) = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} \qquad \text{Welch: df} = \frac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1 - 1} + \dfrac{(s_2^2/n_2)^2}{n_2 - 1}}$$

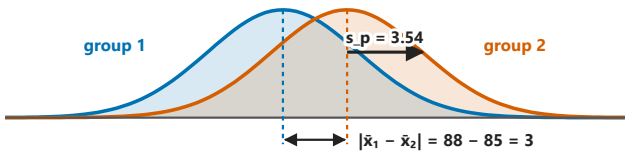


H_0: $\mu_1 = \mu_2$ (difference 0). Small samples → t with Welch df; large → z. $SE_{\text{diff}} = \sqrt{SE_1^2 + SE_2^2}$: two sources of chance, so bigger than either.

Cohen's d — effect size

11.2

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_p}, \qquad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$



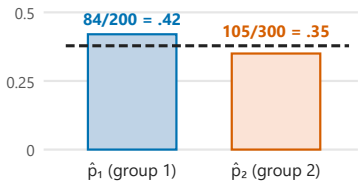
d = 3 / 3.54 ≈ 0.85 → large (0.2 small · 0.5 medium · 0.8 large)

Equal sample sizes: $s_p = \sqrt{(s_1^2 + s_2^2)/2}$. The test asks **whether** there is a difference (÷ SE, shrinks with n); d asks **how big** (÷ SD, doesn't). Significant ≠ big.

Two proportions

11.3

$$\hat{p}_1 = \frac{x_1}{n_1}, \quad \hat{p}_2 = \frac{x_2}{n_2}, \quad \bar{p} = \frac{x_1 + x_2}{n_1 + n_2}, \qquad z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$



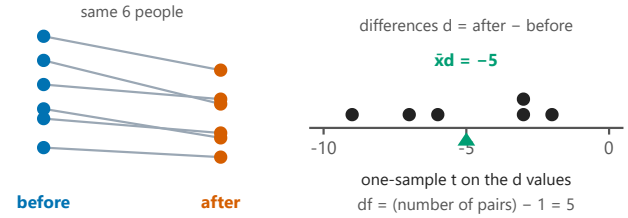
H_0: $p_1 = p_2$
pool (dashed line):
$\bar{p} = (x_1 + x_2)/(n_1 + n_2)$
= 189/500 = 0.378
one shared $\bar{p}$ goes into the SE

Pool because H_0 says the rates are equal without saying what they equal. "Fail to reject" = no significant difference found, not "proven equal".

Matched / paired samples

11.4

$$\bar{x}_d = \frac{\sum x_d}{n}, \quad s_d = \sqrt{\frac{\sum (x_d - \bar{x}_d)^2}{n - 1}}, \quad t = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}}, \quad \text{CI: } \bar{x}_d \pm t\frac{s_d}{\sqrt{n}}$$



Pairing removes person-to-person variation → smaller SE → real effects easier to detect. Match H_a to the claim ("reduces" → $\mu_d < 0$ when d = after − before).

12 The chi-square distribution

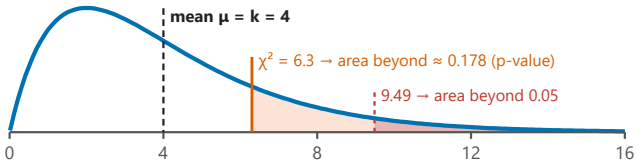
The chi-square distribution 12.1

$$\chi^2_k = Z_1^2 + Z_2^2 + \cdots + Z_k^2 \quad (Z_i \text{ independent standard normal})$$

$$f(x) = \frac{x^{k/2-1}\, e^{-x/2}}{2^{k/2}\,\Gamma\left(\frac{k}{2}\right)}, \quad x > 0 \qquad \mu = k, \quad \sigma^2 = 2k, \quad \sigma = \sqrt{2k}$$

$$\text{df} = 4\text{: } f(x) = \frac{x\, e^{-x/2}}{4}, \qquad P(\chi^2 > x) = e^{-x/2}\left(1 + \frac{x}{2}\right)$$

more df → curve moves right and spreads (σ = √2k)

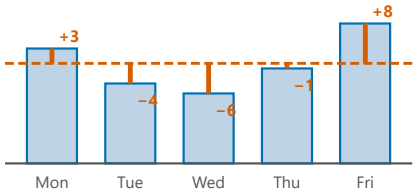


df = 4: f(x) = x·e^(−x/2) / 4, peak at x = k − 2 = 2, mean at k = 4

k = df. $\Gamma(n) = (n-1)!$ for whole numbers, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Only $x > 0$ because it is a sum of squares, and it averages k when H_0 is true. In practice read areas from the χ^2 table.

Goodness of fit 12.1

$$E = np, \qquad \chi^2 = \sum \frac{(O-E)^2}{E}, \qquad \text{df} = k - 1 \quad (k = \text{categories})$$

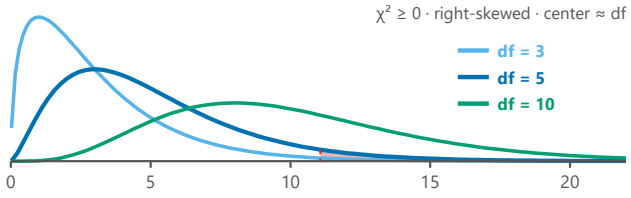


O = observed (bars)
E = n·p = 20 (dashed)
gaps O − E sum to 0
χ² = Σ (O − E)² / E
= 126 / 20 = 6.3
df = k − 1 = 4

Always **right-tailed**: large χ^2 → the data don't fit the claimed distribution → reject H_0.

Test of one variance / SD 12.2

$$\chi^2 = \frac{(n-1)\, s^2}{\sigma_0^2}, \qquad \text{df} = n - 1$$



red tail: α = 0.05 beyond the critical value 11.07 (df = 5)

If H_0 is true, $s^2/\sigma_0^2 \approx 1$ so $\chi^2 \approx$ df. Tables list **right-tail** areas: for a left-tailed test look up $1 - \alpha$ and reject only if χ^2 is **below** it.

Test of independence (contingency table) 12.3

$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}, \qquad \chi^2 = \sum_{\text{all cells}} \frac{(O-E)^2}{E}$$

$$\text{df} = (r-1)(c-1), \quad r = \text{rows}, \; c = \text{columns (totals not counted)}$$

	C1	C2	C3	total
R1	30 E = 40	45	25	100
R2	50	35	15	100
total	80	80	40	200

E(R1, C1)
= row × col / grand
= 100 × 80 / 200 = 40
df = (r − 1)(c − 1)
= (2 − 1)(3 − 1) = 2

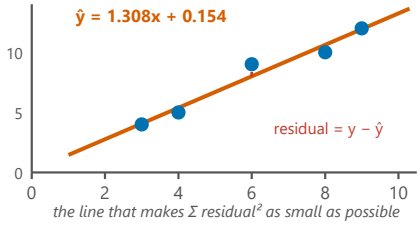
H_0: the two variables are independent

Under independence every row has the same split as the whole table. Large χ^2 → reject independence → the variables are related.

13 Regression and correlation

Least-squares line 13.1

$$\hat{y} = mx + b, \qquad m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}, \qquad b = \frac{\sum y - m\sum x}{n}$$



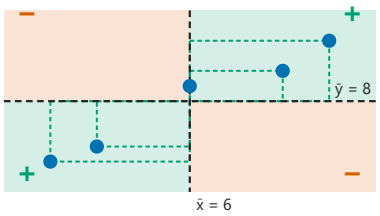
the line that makes Σ residual² as small as possible

n = 5, Σx = 30, Σy = 40
Σxy = 274, Σx² = 206
m = (5·274 − 30·40)
/ (5·206 − 30²)
= 170/130 ≈ 1.308
b = (40 − m·30)/5 ≈ 0.154

Table columns x, y, xy, x^2 (and y^2 for r). Don't round m before computing b. Excel: `SLOPE`, `INTERCEPT`.

Covariance 13.3

$$\text{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1} \qquad \text{(population: } \div N\text{)}$$



each box = (x − x̄)(y − ȳ)
12 + 6 + 0 + 4 + 12
Σ = 34
cov = 34 / (5 − 1) = 8.5
all points in + corners
→ positive relationship

The **sign** gives the direction; the size depends on the units. Each deviation column sums to 0 — a handy check.

Correlation coefficient r 13.2 · 13.4

$$r = \frac{\text{cov}(x,y)}{s_x\, s_y} = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - \left(\sum x\right)^2\right]\left[n\sum y^2 - \left(\sum y\right)^2\right]}}$$



r = 0.86
strong positive

r = −0.03
no linear link

r = −0.94
strong negative

$-1 \le r \le 1$. Sign = direction, size = strength of the **linear** link. $r \approx 0$ can still hide a curved relation. Example data: $r = 8.5/(2.55 \times 3.39) \approx 0.983$. Correlation is not causation.

★ Which formula do I need?

Estimating (confidence intervals) and testing (hypothesis tests)

situation	confidence interval	test statistic	distribution / df	lesson
one mean, σ known (or $n \geq 30$)	$\bar{x} \pm z\dfrac{\sigma}{\sqrt{n}}$	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	normal	9.1 · 10.4
one mean, σ unknown, $n < 30$	$\bar{x} \pm t\dfrac{s}{\sqrt{n}}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	t, df $= n - 1$	9.3 · 10.5
one proportion	$\hat{p} \pm z\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0/n}}$	normal ($n\hat{p}\hat{q} \geq 10$)	9.4 · 10.7
two independent means	$(\bar{x}_1 - \bar{x}_2) \pm t\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$\dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$	t (Welch df) or z	11.1
two proportions	—	$\dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\bar{p}(1-\bar{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$	normal, pooled $\bar{p}$	11.3
paired (before / after)	$\bar{x}_d \pm t\dfrac{s_d}{\sqrt{n}}$	$t = \dfrac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}}$	t, df $= n_{\text{pairs}} - 1$	11.4
one variance or SD	—	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$	χ^2, df $= n - 1$	12.2
counts vs a claimed split	—	$\chi^2 = \sum \dfrac{(O-E)^2}{E}$, $E = np$	χ^2, df $= k - 1$, right tail	12.1
two categorical variables	—	χ^2 with $E = \dfrac{\text{row} \times \text{col}}{\text{grand}}$	χ^2, df $= (r-1)(c-1)$	12.3
how big is a difference?	Cohen's $d = \lvert\bar{x}_1 - \bar{x}_2\rvert / s_p$ — 0.2 small · 0.5 medium · 0.8 large			11.2
planning a study	$n = z^2\sigma^2/E^2$ (mean) · $n = z^2\hat{p}(1-\hat{p})/E^2$ (proportion, $\hat{p} = 0.5$ if unknown) — round up			9.5